



Making Decisions

BITS Pilani
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Learning Outcomes

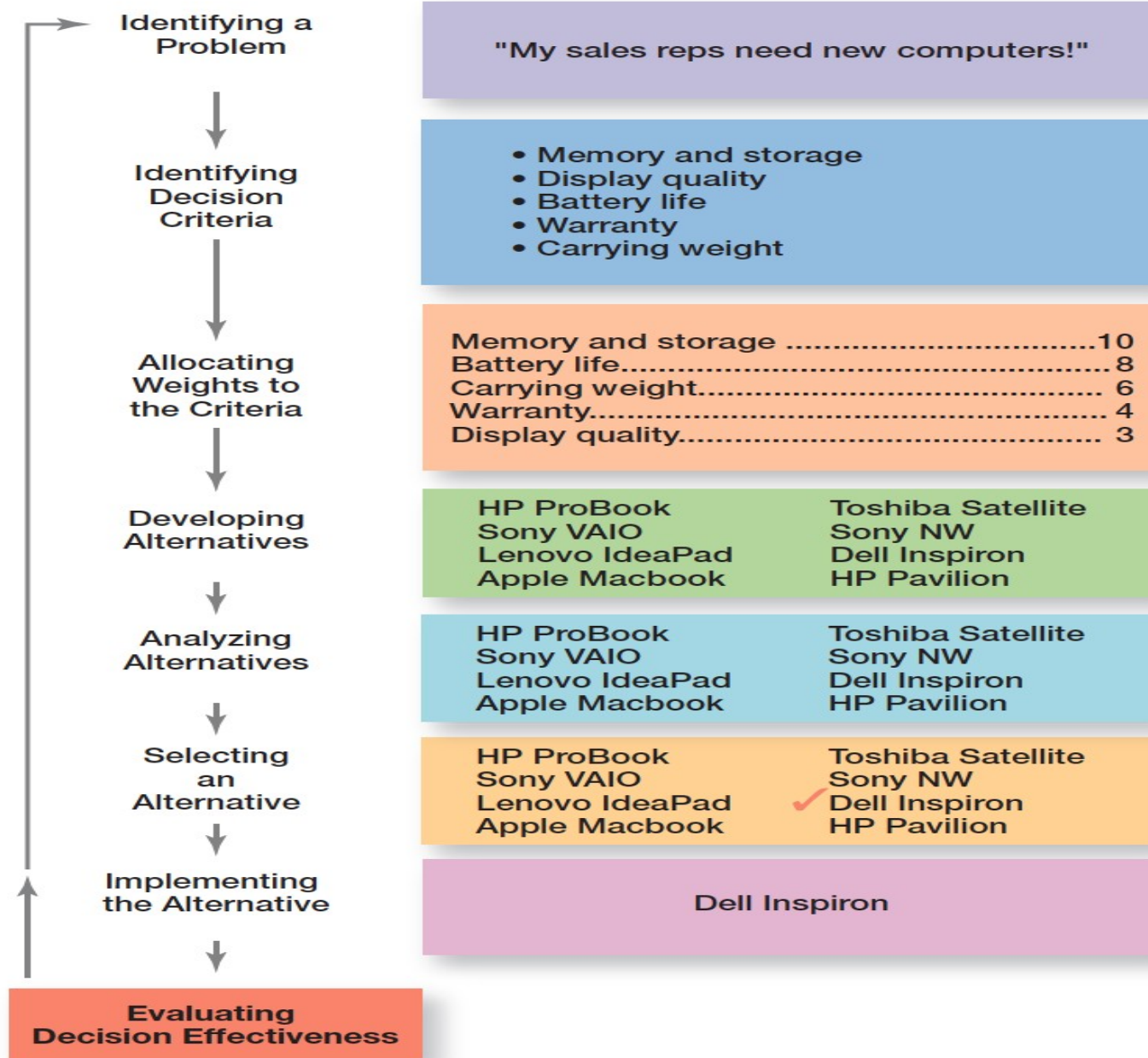
- **Eight steps** in the decision-making process
 - Four **ways or perspectives** of how managers make decisions.
 - Decisions **types**
 - Decision-making **conditions**
 - Decision-making **Styles**
 - How **biases** affect decision making
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What is a Decision?

Decision—a choice among two or more alternatives



Decision Making Process





Important Decision Criteria

Criterion	Weight
Memory and storage	10
Battery life	8
Carrying weight	6
Warranty	4
Display quality	3



Possible Alternatives

Laptop	Memory and Storage	Battery Life	Carrying Weight	Warranty	Display Quality
HP ProBook	10	3	10	8	5
Lenovo IdeaPad	8	5	7	10	10
Apple MacBook	8	7	7	8	7
Toshiba Satellite	7	8	7	8	7
Apple MacBook Air	8	3	6	10	8
Dell Inspiron	10	7	8	6	7
HP Pavilion	4	10	4	8	10



Evaluation of Alternatives

Laptop	Memory and Storage	Battery Life	Carrying Weight	Warranty	Display Quality	Total
HP ProBook	100	24	60	32	15	231
Lenovo IdeaPad	80	40	42	40	30	232
Apple MacBook	80	56	42	32	21	231
Toshiba Satellite	70	64	42	32	21	229
Apple MacBook Air	80	24	36	40	24	204
Dell Inspiron	100	56	48	24	21	249
HP Pavilion	40	80	24	32	30	206

Decisions Managers May Make



Planning

- What are the organization's long-term objectives?
- What strategies will best achieve those objectives?
- What should the organization's short-term objectives be?
- How difficult should individual goals be?

Organizing

- How many employees should I have report directly to me?
- How much centralization should there be in an organization?
- How should jobs be designed?
- When should the organization implement a different structure?

Leading

- How do I handle employees that appear to be unmotivated?
- What is the most effective leadership style in a given situation?
- How will a specific change affect worker productivity?
- When is the right time to stimulate conflict?

Controlling

- What activities in the organization need to be controlled?
- How should those activities be controlled?
- When is a performance deviation significant?
- What type of management information system should the organization have?

Four perspectives

how managers make decisions



How you make decision....

You are founder & CEO of an Electric Car Manufacturing startup in the process to establish your operations to cater India and UK markets.

What all Decisions you may need to take?.....

- 1.Manufacture or Buy Components?
- 2.Which Supplier(s) to bring on board?

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How will you make decision here?.....

Perspective 1:

Making Decisions: Rationality



We assume that managers will **use Rational Decision-Making**, will make **logical** and **consistent choices** to **maximize value** (after assessing all possible alternatives). Managers have all sorts of tools and techniques to help them be rational decision makers.

•Assumptions of rationality:

- Rational decision maker is logical and objective
 - Problem faced is clear and unambiguous
 - Decision maker would have clear, specific goal and be aware of all alternatives and consequences
 - The alternative that maximizes achieving this goal will be selected
 - Decisions are made in the best interest of the organization
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Perspective 2:

Decision Making: Bounded Rationality



Bounded rationality: Managers make decisions rationally, but are limited (bounded) by their ability to process information.

- Managers **Satisfice**, rather than maximize: accepting solutions that are “**good enough**”.

(Important to keep in mind that decision making is also influenced by the *organization's culture, internal politics, power considerations* and by a phenomenon called *escalation of commitment*)

- **Escalation of commitment:** an increased commitment to a previous decision despite evidence it may have been wrong.

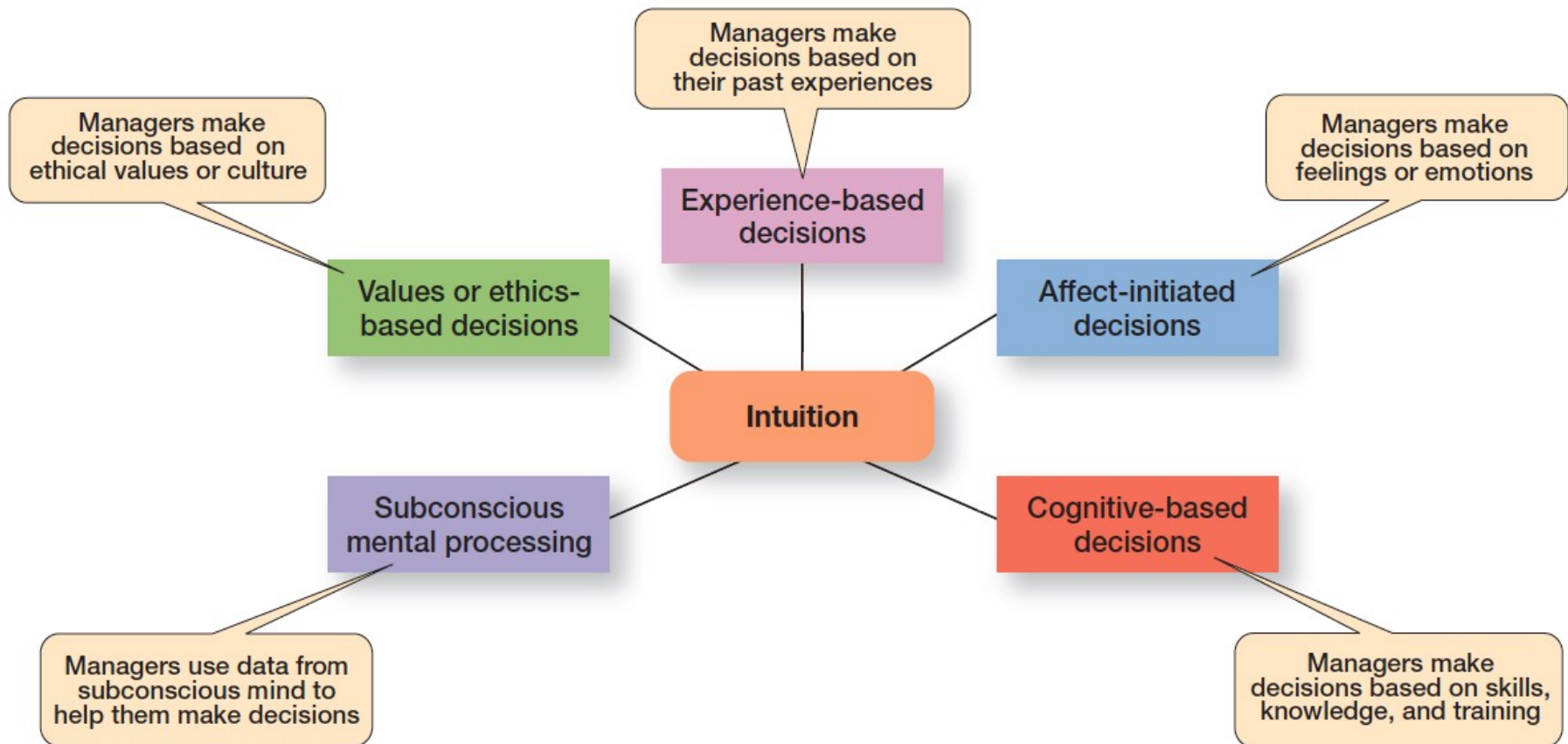
Perspective 3:

Making Decisions: The Role of Intuition



- **Intuitive decision-making:** making decisions on the **basis of experience, feelings, and accumulated judgment.**
 - Intuitive decision making can **complement** both rational and bounded rational decision making.
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What is Intuition?



Shows the five different aspects of intuition identified by researchers studying managers' use of intuitive decision-making.

Perspective 4:

Evidence-Based Management



Evidence-based management (EBMgt): the systematic use of the best available evidence to improve management practice. Use **relevant** and **reliable evidence**.

Four essential elements of EBMgt are:

1. The decision maker's expertise and judgement
 2. External evidence that's been evaluated by the decision maker
 3. Opinions, preferences, and values of those who have a stake in the decision
 4. Relevant organizational (internal) factors such as context, circumstances, and organizational members.
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Common Decision-Making Biases





Decision-Making Biases and Errors (1 of 2)

- **Overconfidence Bias:** holding **unrealistically positive views** of oneself and one's performance
- **Immediate Gratification Bias:** choosing alternatives that **offer immediate rewards** and **avoid immediate costs**
- **Anchoring Effect:** fixating on **initial information** and **ignoring** subsequent information
- **Selective Perception Bias:** selecting, organizing and interpreting events based on the **decision maker's biased perceptions**
- **Confirmation Bias:** seeking out information that reaffirms past choices while **discounting contradictory information**
- **Framing Bias:** selecting and **highlighting certain aspects** of a situation while ignoring other aspects



Decision-Making Biases and Errors (2 of 2)

- **Availability Bias:** losing decision-making objectivity by focusing on the **most recent events**
- **Representation Bias:** **drawing analogies** and seeing identical situations when none exist
- **Randomness Bias:** creating unfounded meaning out of **random events**
- **Sunk Costs Errors:** forgetting that current actions cannot influence past events and relate only to future consequences
- **Self-serving Bias:** taking quick credit for successes and **blaming outside factors** for failures



Types of Decisions

Structured problems: straightforward, familiar, and easily defined problems.

- **Programmed decisions:** repetitive decisions that can be handled by a routine approach.

Types of Programmed Decisions

- **Procedure:** a series of sequential steps used to respond to a well-structured problem
 - **Rule:** an explicit statement that tells managers what can or cannot be done
 - **Policy:** a guideline for making decisions
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Types of Decisions

- **Unstructured problems:** problems that are new or unusual and for which information is ambiguous or incomplete
 - **Nonprogrammed decisions:** unique and nonrecurring and involve custom made solutions
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Decision-Making Conditions

- **Certainty:** a situation in which a manager can make accurate decisions because all outcomes are known.
 - **Risk:** a situation in which the decision maker is able to estimate the likelihood of certain outcomes.
 - **Uncertainty:** a situation in which a decision maker has neither certainty nor reasonable probability estimates available.
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Managing Risk

- Managers can use historical data or secondary information to assign probabilities to different alternatives
 - This is used to calculate expected value—the expected return from each possible outcome—by multiplying expected revenue by the probability of each alternative
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Example: Expected Value

On the basis of last 10 years data

Event	Expected Revenues	Probability	Expected Value of Each Alternative
Heavy snowfall	\$850,000	0.3	\$255,000
Normal snowfall	\$725,000	0.5	\$362,000
Light snowfall	\$350,000	0.2	\$70,000
Total expected revenue:			\$687,500

Uncertainty

Under uncertainty, the choice of alternatives is influenced by the limited amount of available information and by psychological orientation of the decision maker.

- An **optimistic manager** will follow a *maximax* choice (maximizing the maximum possible payoff).
 - A **pessimist** will follow a *maximin* choice (maximizing the minimum possible payoff).
 - A manager who desires to minimize his maximum “**regret**” will opt for a *minimax* choice.
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Example: Decision making under uncertainty

Payoff Matrix

Visa Marketing Strategy (in millions of dollars)	MasterCard CA 1	MasterCard CA 2	MasterCard CA 3
Strategy 1	13	14	11
Strategy 2	9	15	18
Strategy 3	24	21	15
Strategy 4	18	14	28

Regret Matrix

Visa Marketing Strategy (in millions of dollars)	MasterCard CA 1	MasterCard CA 2	MasterCard CA 3
Strategy 1	11	7	17
Strategy 2	15	6	10
Strategy 3	0	0	13
Strategy 4	6	7	0

shows the regret for each category, which is found by subtracting all possible payoffs in each category from the maximum possible payoff for each given event, in this case for each competitive action.



Decision-making Styles

The **way a person approaches** to decision making is likely affected by his or her **thinking style**. Thinking style reflects two things:

- 1) the **source of information** you tend to use (external data and facts OR internal sources such as feelings and intuition)
- 2) whether you **process** that information in a linear way (rational, logical, analytical) OR a nonlinear way (intuitive, creative, insightful)

Linear thinking Style: a person's preference for using **external data and facts** and processing this information through **rational, logical thinking** to guide decisions and actions.

Nonlinear thinking Style: a preference for **internal sources** of information (**feelings and intuition**) and processing this information with **internal insights, feelings, and hunches** to guide decisions and actions.



Guidelines for Making Effective Decisions

- Understand **cultural** differences
- Create **standards** for good decision making
- Know when it's time to **call it quits**
- Use an **effective decision-making** process
- Develop your ability to **think clearly**

Design thinking: **approaching** management **problems** as designers approach design problems. **Understanding the problem first is the key**

Big data: the vast amount of quantifiable data that can be analyzed by highly sophisticated data processing



Planning Work Activities

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What is Planning?

- **Planning:** **primary** management function that involves setting **goals**, establishing **strategies** for achieving those goals, and developing **plans** to integrate and coordinate work activities.

It involves two important aspects:

- **Goals (objectives):** desired outcomes or targets
 - **Plans:** documents that outline how goals are going to be met
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Why Do Managers Plan?

- Provides direction
 - Reduces uncertainty
 - Minimizes waste and redundancy
 - Establishes the goals and standards for controlling
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Approaches to Setting Goals

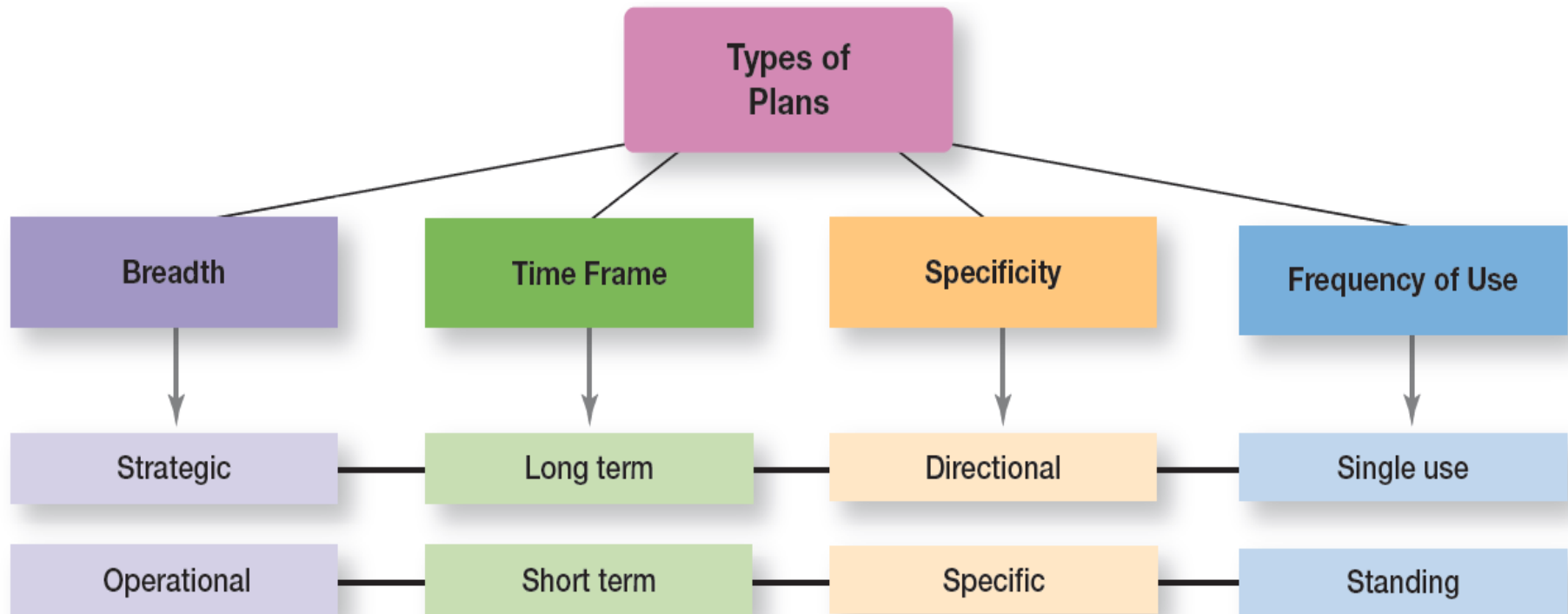
- **Traditional goal-setting:** an approach to setting goals in which **top managers set goals** that then flow down through the organization and become sub-goals for each organizational area
 - **Means-ends chain:** an integrated network of goals in which the accomplishment of goals at one level serves as the means for achieving the goals, or ends, at the next level
 - **Management by objectives (MBO):** a process of setting **mutually agreed upon goals** and using those goals to evaluate employee performance
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The Downside of Traditional Goal-Setting



Exhibit 8-2 illustrates what can happen as the goals make their way down from the top of the organization to lower levels.

Types of Plans





Strategic and Operational Plans

- **Strategic plans:** plans that apply to the entire organization and establish the organization's overall goals
 - **Operational plans:** plans that encompass a particular operational area of the organization
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Long-term and Short-term Plans

- **Long-term plans:** plans with a time frame beyond three years
- **Short-term plans:** plans covering one year or less



Specific and Directional Plans

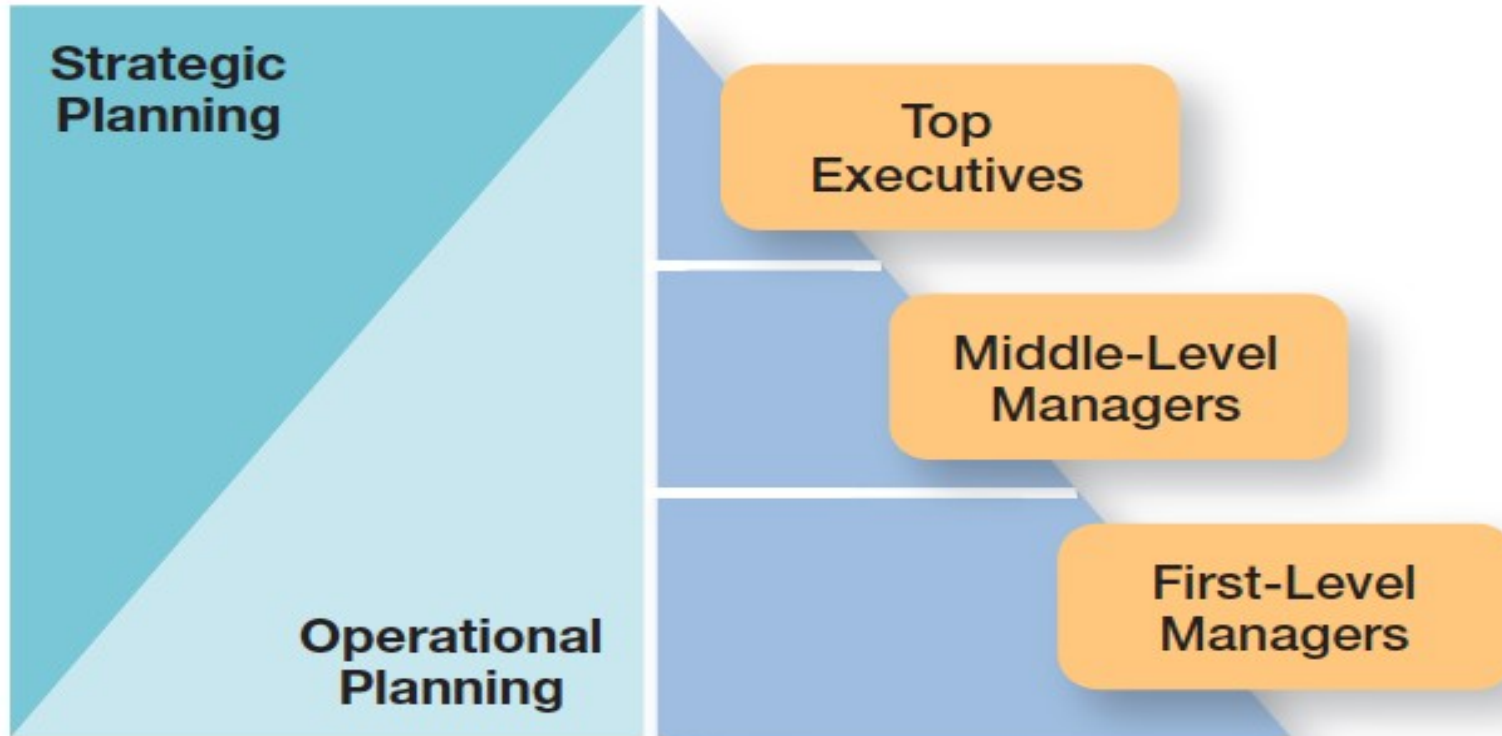
- **Specific plans:** plans that are clearly defined and leave no room for interpretation
 - **Directional plans:** plans that are flexible and set out general guidelines
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Single-use and Standing Plans

- **Single-use plan:** a one-time plan specifically designed to meet the needs of a unique situation
 - **Standing plan:** ongoing plan that provide guidance for activities performed repeatedly
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Planning and Organizational Level



Two Approaches to Developing Plans:

1. Top level managers assisted by **Formal Planning department**
2. Involve more organizational members in the process

How Can Managers Plan Effectively in Dynamic Environments?



- Develop plans that are specific but flexible
- Keep planning even when the environment is uncertain
- Allow lower organizational levels to set goals and develop plans





Thank you